# CHEMISTRY

## SECTION II

## TIME – 1 HOUR AND 45 MINUTES

**Directions:**

Section II has 7 questions and lasts 1 hour and 45 minutes.

You may use the available paper for scratch work and planning, but you must write your answers in the free-response booklet. Label parts (e.g., A, B, C) and sub-parts (e.g., i, ii, iii) as needed. Use a pencil or a pen with black or dark blue ink to write your responses.

For each question, show your work for each part in the space provided for that part in the free-response booklet. Examples and equations may be included in your responses where appropriate. For calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Pay attention to significant figures.

A calculator is allowed in this section. You may use a handheld four-function, scientific, or graphing calculator, or the calculator available in this application. Reference information, including a periodic table and lists of equations and constants, is available in this application and can be accessed throughout the exam.

You may pace yourself as you answer the questions in this section, or you may use these optional timing recommendations:

Questions 1–3 are long free-response questions. It is suggested that you spend about 23 minutes on each of those questions. Questions 4–7 are short free-response questions. It is suggested that you spend about 9 minutes on each of those questions.

You can go back and forth between questions in this section until time expires. The clock will turn red when 5 minutes remain—**the proctor will not give you any time updates or warnings.**

Note: This exam was originally administered digitally. It is presented here in a format optimized for teacher and student use in the classroom.

During the AP Exam administration, students have access to reference information. To see the reference information for this course, please visit AP Central:
https://apcentral.collegeboard.org/courses/ap-chemistry/exam

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**2.** Answer the following questions about ascorbic acid (vitamin C).

**A.** A student combusts a sample of ascorbic acid, $C_xH_yO_x$, to determine its chemical composition. The only products of the reaction are 0.2400 mol of $CO_2$ and 2.883 g of $H_2O$.

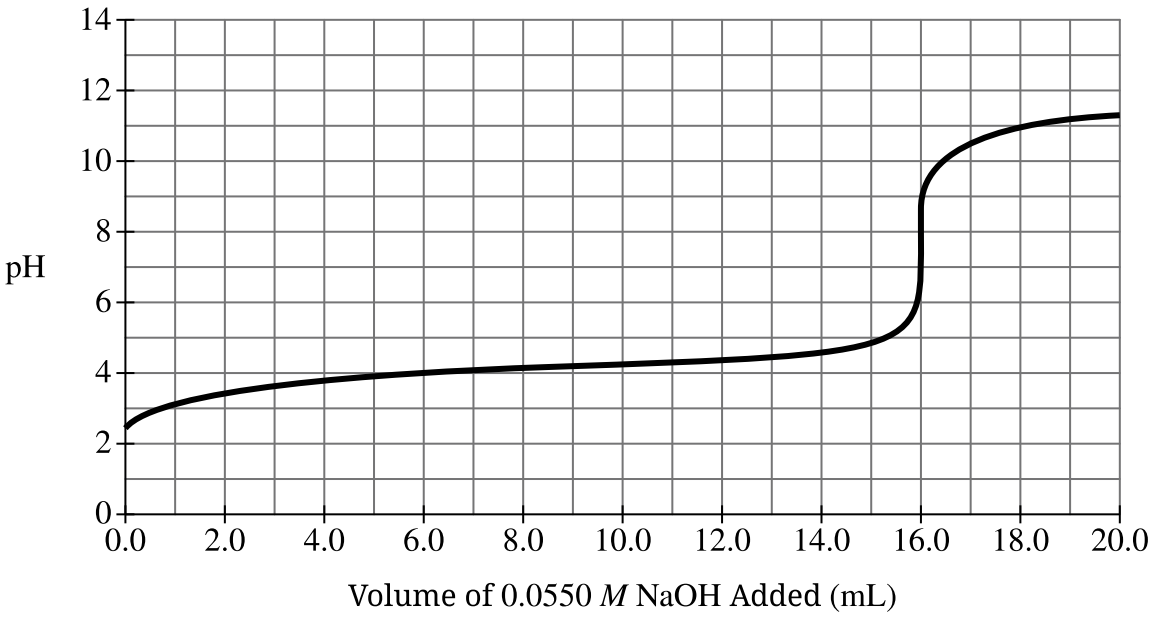
i. Calculate the number of moles of $H_2O$ produced.

ii. The mole ratio of carbon (C) to oxygen (O) is 1:1 in ascorbic acid. Based on this information and your answer to part A (i), determine the empirical formula of ascorbic acid.

**B.** Ascorbic acid, $HAsc(aq)$, acts as a weak acid, as shown in the equation.

$$HAsc(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + Asc^-(aq)$$

The following titration curve was produced when a 10.0 mL sample of $HAsc(aq)$ was titrated using 0.0550 $M$ $NaOH(aq)$.




i. Calculate the molar concentration of the ascorbic acid solution.

ii. From the titration curve, determine the approximate $pK_a$ of ascorbic acid.

iii. What is the value of the ratio $\frac{[Asc^-]}{[HAsc]}$ when the pH of the solution is 4.7?

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**C.** Dehydroascorbic acid (DHAsc) can be produced by reacting ascorbic acid with the triiodide ion, $I_3^-$, as represented by the following equation.

$$\text{HAsc} + I_3^- \rightarrow \text{DHAsc} + 3\ I^- + 2\ H^+$$

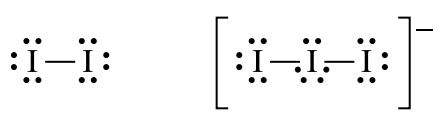
The student runs three trials of the reaction with different initial concentrations of HAsc and $I_3^-$, producing the following data.

| Trial | [HAsc] $(M)$ | $[I_3^-]$ $(M)$ | Initial Rate of DHAsc Formation $(M/\text{s})$ |
|---|---|---|---|
| 1 | 0.450 | 1.200 | $2.457\times10^{-4}$ |
| 2 | 0.450 | 0.600 | $1.229\times10^{-4}$ |
| 3 | 0.900 | 1.200 | $4.914\times10^{-4}$ |

i. The rate law for the reaction is $rate = k[\text{HAsc}][I_3^-]$. Explain how the data in the table support the conclusion that the reaction is first order with respect to $[\text{HAsc}]$.

ii. Calculate the value of the rate constant, $k$, for the reaction. Include units with your answer.

**D.** The triiodide ion, $I_3^-$, is significantly more soluble in water than elemental iodine, $I_2$, is. Identify an intermolecular force between $I_3^-$ and water that is **not** present between $I_2$ and water, which could explain the difference in solubility. Lewis diagrams for $I_2$ and $I_3^-$ are provided.



$I_2$ $\qquad$ $I_3^-$

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